

Data Model and API Reference

Storage schema, indexes, and HTTP API for the Medical FAQ project.

1. Storage Overview

Two stores are used. MongoDB holds chat sessions, the FAQ store, and the cluster history. Qdrant holds dense embeddings used by the retrieval service. The split keeps write-heavy chat logs on a general-purpose database and latency-sensitive search on a vector DB.

Storage Schema

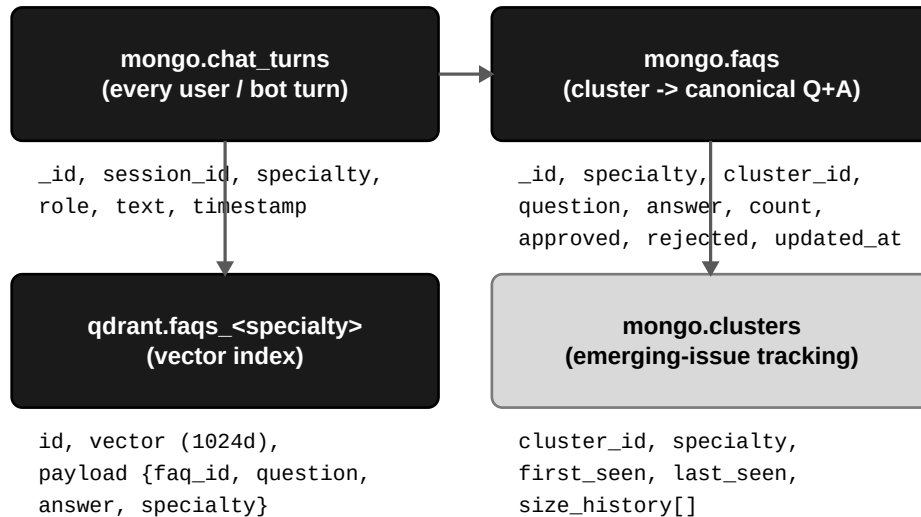


Fig 1. MongoDB collections (top) and Qdrant collection (bottom-left).

2. MongoDB Collections

2.1 chat_turns

Append-only log of every turn in every session, user and bot. Written by the FastAPI `/chat` endpoint and consumed by the pipeline ingest stage.

Field	Type	Notes
<code>_id</code>	ObjectId	Mongo auto-id.
<code>session_id</code>	string	Client-side UUID, groups one user conversation.
<code>specialty</code>	string	One of: radiology, physiotherapy, cardiovascular.
<code>role</code>	string	user or bot.
<code>text</code>	string	Already PHI-scrubbed at write time.
<code>timestamp</code>	datetime	UTC. Indexed for ingest queries.

2.2 faqs

One document per cluster. Pipeline upserts on `(specialty, cluster_id)`. Admin dashboard flips `approved` to true. Public FAQ endpoint reads only approved entries.

Field	Type	Notes
<code>_id</code>	ObjectId	Mongo auto-id.

specialty	string	Specialty tag.
cluster_id	int	Stable id from HDBSCAN result.
question	string	Canonical question (closest to cluster centroid).
answer	string	Extractively selected agent reply, polished.
count	int	Cluster size at publish time.
approved	bool	Clinician approval flag. Defaults false.
rejected	bool	Set true if a clinician rejected this candidate.
updated_at	datetime	Last pipeline run that touched this row.

2.3 clusters

Tracks size over time per cluster so the emerging-issue detector can spot fast growth. Written at the end of the cluster stage.

Field	Type	Notes
cluster_id	int	Stable id.
specialty	string	Specialty tag.
first_seen	datetime	First time this cluster appeared.
last_seen	datetime	Most recent pipeline run that produced it.
size_history	list	Daily sizes for the trailing 14 days.

3. Qdrant Collection

One collection per specialty: `faqs_radiology`, `faqs_physiotherapy`, `faqs_cardiovascular`. Vector dimension is 1024 (BGE-M3). Distance is cosine, with vectors normalised at write time so dot-product is enough at query time.

Setting	Value
Vector dimension	1024
Distance	Cosine
Quantization	Off for the prototype, scalar later if needed.
Payload schema	faq_id, specialty, question, answer
Write path	Pipeline publish stage. Upserts one point per FAQ.
Read path	FastAPI retrieval service on /chat.

4. Indexes

Collection	Index	Reason
chat_turns	(specialty, timestamp)	Pipeline ingest pulls new turns by time, filtered by specialty.
chat_turns	session_id	Group turns into sessions inside the ingest stage.
faqs	(specialty, approved)	Public FAQ endpoint reads only approved rows per specialty.
faqs	(specialty, cluster_id)	Pipeline upsert key. Unique.
clusters	(specialty, last_seen)	Emerging-issue detector scans recent clusters.

5. API Overview

FastAPI exposes one health check plus three routers: /chat, /faq, and /admin. All responses are JSON. The Next.js frontend is the primary consumer; Swagger UI is on /docs for manual testing.

API Endpoint Hierarchy

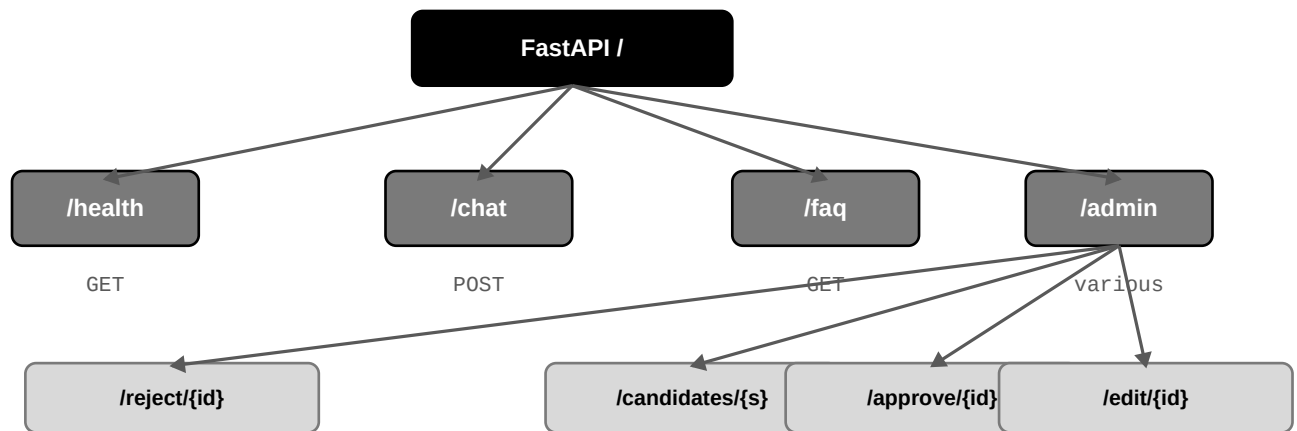


Fig 2. API endpoint hierarchy.

6. POST /chat

Take a user message, log it, return the closest stored answer.

Request body

Field	Type	Required	Notes
session_id	string	yes	Client-side UUID.
specialty	string	yes	One of the three specialties.
text	string	yes	User message. PHI scrubbed at write time.

Response (200 OK)

Field	Type	Notes
answer	string	Selected answer from the FAQ index, or a polite fallback.
matched_faq_id	string null	FAQ id when confidence is above threshold.
confidence	float	Cosine similarity of the top retrieval hit.

7. GET /faq/{specialty}

Return approved FAQs for the specialty, ranked by cluster size. Pagination is via the limit query param (default 50).

Field	Type	Notes
id	string	Mongo _id as a hex string.
specialty	string	Echoed back.
question	string	Canonical question.
answer	string	Polished extractive answer.
count	int	Cluster size.

updated_at	datetime (ISO)	Last pipeline run.
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8. Admin endpoints

Endpoint	Method	Body	Purpose
/admin/candidates/{specialty}	GET	-	List candidate FAQs awaiting review.
/admin/approve/{faq_id}	POST	-	Mark FAQ as approved (visible on public FAQ page).
/admin/edit/{faq_id}	POST	{question?, answer?}	Edit one or both of the FAQ fields.
/admin/reject/{faq_id}	POST	-	Mark FAQ as rejected. Never auto-republished.

Error model

Code	When	Body
400	Bad input (missing fields, bad id format)	{detail: 'Invalid FAQ id'}
404	FAQ not found	{detail: 'FAQ not found'}
422	Pydantic validation failure	FastAPI default validation envelope
500	Service error (Mongo down)	{detail: 'Internal Server Error'}

9. Sample Requests

POST /chat

```
curl -X POST http://localhost:8000/chat -H 'Content-Type: application/json' -d '{"session_id":"abc","specialty":"radiology","text":"do i need to fast before mri?"}'
```

GET /faq/radiology

```
curl http://localhost:8000/faq/radiology
```

POST /admin/approve/{id}

```
curl -X POST http://localhost:8000/admin/approve/65b1c2d3e4f5a6b7c8d9e0f1
```